LESSON 316 (1.0 AATD)

Lesson Objective
During this lesson, intersection, and GPS holding patterns will be introduced.

Briefing
- Review homework assigned in the previous lesson.
- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Student questions.

Lesson Content (Simulator)
System Power Up
- Student Tells, Student Does.
  - Have the student start the simulator using the checklist and assist.
  - Ensure that the student checks GPS database currency.
    - Half-left, half-up, flag out of view.
  - Ensure that there are no red 'x's.
  - Have the student start the engine using the checklist.

Instrument Cockpit Check
- Student Tells, Student Does.
  - Have the student read each item from the checklist out loud.
  - Demonstrate to the student what needs to be done for each item.
  - Discipline the student to set bugs, bearing pointers, and any auxiliary features.

Before Taxi Checks
- Student Tells, Student Does.
  - Airspeed indicator (PFD and standby): zero.
  - Attitude indicator (PFD): pitch/bank deviations +/- 5 degrees.
  - Standby attitude indicator: discuss standby battery check and adjust the miniature aircraft (cannot be performed in the ATD).
  - Altimeter (PFD and standby): +/- 75 feet of station elevation.
  - Vertical speed indicator (PFD): reads 0 or use current indication.
  - Trim coordinator: wings level or no vector when still; inclinometer centered when still; skids when turning.
  - HSI: matches compass and turns appropriately.
  - Magnetic compass: indicates heading properly, no bubbles, card present, and swings freely.

Give the student a departure via radar vectors. The instructor simulates ATC. Have the student switch to the appropriate frequencies throughout the simulator lesson.

<u>**Note**</u>:

- For this lesson do not pause the sim when giving holding instructions. This will create a similar environment for when the student will be flying.
- Set a moderate wind of around 10-20 knots to evaluate and emphasize the need for wind and time corrections.
- You can use the examples given below or choose any fix in the vicinity.

GPS Holding Patterns:

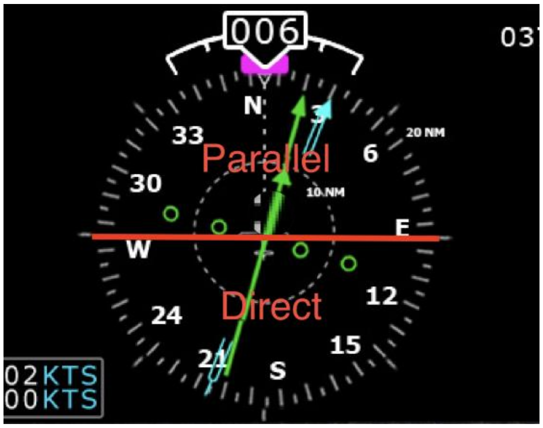
- Standard hold:
  <u>Student Tells, Student Does.</u>
  - Clear the student direct to a GPS fix (e.g. VALKA, TPSTR, SMERE, MLBSO, etc.).
  - If the student is unable to spell the fix, encourage them to ask for the spelling or use the nearest page to find the fix.
  - Give the student a standard holding instruction
    - *Hold NE of BEZLE on the 040 course, EFC 15 min.*
  - Ensure that the student repeats and understands the instructions.
  - Ensure that the student can determine the correct entry procedure.
  - Verify the completion of the 6 Ts (note that the order may change slightly depending on entry):
    - **T**urn - upon crossing fix, turn to the outbound heading.
      - This will be the given bearing.
    - **T**alk - advise ATC established in the hold.
    - **T**ime - start the timer at **wings level** or **flag flip**, **whichever happens last.**
    - **T**wist - **press OBS** and twist to the inbound course.
      - This will be the reciprocal of the given bearing.
    - **T**rack - after one minute or the holding distance, turn inbound and track the inbound course.
    - **T**hrottle - adjust throttle for 90 kts.
  - Ensure the student is making proper wind and time corrections.
    - Wind correction: once established inbound for the first time, note the difference between the current heading and the ground track on the PFD. This difference x 3 should be applied into the direction of the wind to the outbound heading.
    - Time correction: time each full inbound leg from wings level until reaching the fix. Note the difference between this time and one minute. Add or subtract this difference to the time of each outbound leg.
      - Note: do not time the first inbound leg as a part of the entry. This correction only applies to full inbound legs once established in the hold.
  - While holding, ensure the student is continuing with the radial scan.
  - Repeat procedures with different instructions until the student is proficient.

- Non-standard hold:
  Student Tells, Student Does.
  - Clear the student direct to a GPS fix (e.g. VALKA, TPSTR, SMERE, MLBSO, etc.).
  - If the student is unable to spell the fix, encourage them to ask for the spelling or use the nearest page to find the fix.
  - Give the student a non-standard holding instruction:
    - *Hold SE of the Valka intersection on the 120 bearing, left turns, EFC 15 mins.*
  - Ensure that the student repeats and understands the instructions.
  - Ensure that the student can determine the correct entry procedure.
  - Verify the completion of the 6 Ts (note that the order may change slightly depending on entry):
    - **T**urn - upon crossing fix, turn to the outbound heading.
      - This will be the given bearing.
    - **T**alk - advise ATC established in the hold.
    - **T**ime - start the timer at **wings level** or **flag flip**, **whichever happens last.**
    - **T**wist - **press OBS** and twist to the inbound course.
      - This will be the reciprocal of the given bearing.
    - **T**rack - after one minute or the holding distance, turn inbound and track the inbound course.
    - **T**hrottle - adjust throttle for 90 kts.
  - Ensure the student is making proper wind and time corrections.
    - Wind correction: once established inbound for the first time, note the difference between the current heading and the ground track on the PFD. This difference x 3 should be applied into the direction of the wind to the outbound heading.
    - Time correction: time each full inbound leg from wings level until reaching the fix. Note the difference between this time and one minute. Add or subtract this difference to the time of each outbound leg.
      - Note: do not time the first inbound leg as a part of the entry. This correction only applies to full inbound legs once established in the hold.
  - While holding, ensure the student is continuing with the radial scan.
  - Repeat procedures with different instructions until the student is proficient.

Intersection Holding Patterns - DME Fix:

- Standard hold:
  Student Tells, Student Does.
  - Give the student instructions to join a VOR radial either inbound or outbound.
  - Give the student a non-standard holding instruction:
    - *Hold N of the 10 DME on present radial, left turns, EFC 15 mins.*
  - Ensure that the student repeats and understands the instructions.
  - Ensure that the student can determine the correct entry procedure.

        - If the given **cardinal direction (N, E, NE, etc.).** falls on the **top** of the CDI, it will be a **parallel** entry.
        - If the given **cardinal direction (N, E, NE, etc.).** falls on the **bottom** of the CDI, it will be a **direct** entry.



    - Verify the completion of the 6 Ts (note that the order may change slightly depending on entry):
        - **T**urn - upon crossing fix, turn to the outbound heading.
        - **T**alk - advise ATC established in the hold.
        - **T**ime - start the timer at wings level - **there is no flag flip on a DME hold.**
        - **T**wist: twist the CDI to the inbound course **if necessary.**
            - Refer to intersection holding patterns in Lesson 314.
            - If **direct**, there will be no need to twist.
            - If **parallel**, twist to the reciprocal of the current CDI course.
        - **T**rack - after one minute or the holding distance, turn inbound and track the inbound course.
        - **T**hrottle - adjust throttle for 90 kts.
    - Ensure the student is making proper wind and time corrections.
        - Wind correction: once established inbound for the first time, note the difference between the current heading and the ground track on the PFD. This difference x 3 should be applied into the direction of the wind to the outbound heading.
        - Time correction: time each full inbound leg from wings level until reaching the fix. Note the difference between this time and one minute. Add or subtract this difference to the time of each outbound leg.
            - Note: do not time the first inbound leg as a part of the entry. This correction only applies to full inbound legs once established in the hold.
    - When holding, ensure the student is continuing with the radial scan.
    - Repeat procedures with different instructions until the student is proficient.

Intersection Holding Patterns - Victor/Tango Airway Intersection

- Standard hold:
- <u>Student Tells, Student Does.</u>
    - Give the student instructions to join a Victor or Tango airway which leads to an

intersection with another airway.
    - Give the student a standard holding instruction:
        - *Hold S of VALKA on T345, EFC 15 mins*
    - Ensure that the student repeats and understands the instructions.
    - Turn on the bearing pointer and set its navigation source to the cross-airway. The fix occurs when the bearing pointer reaches the course of the cross-airway.
    - Ensure that the student can determine the correct entry procedure.
    - Verify the completion of the 6 Ts (note that the order may change slightly depending on entry):
        - **T**urn - upon crossing fix, turn to the outbound heading.
        - **T**alk - advise ATC established in the hold.
        - **T**ime - start the timer at wings level - **there is no flag flip on an intersection hold.**
        - **T**wist: twist the CDI to the inbound course **if necessary.**
            - Refer to intersection holding patterns in Lesson 314.
            - If **direct**, there will be no need to twist.
            - If **parallel**, twist to the reciprocal of the current CDI course.
        - **T**rack - after one minute or the holding distance, turn inbound and track the inbound course.
        - **T**hrottle - adjust throttle for 90 kts.
    - Ensure the student is making proper wind and time corrections.
        - Wind correction: once established inbound for the first time, note the difference between the current heading and the ground track on the PFD. This difference x 3 should be applied into the direction of the wind to the outbound heading.
        - Time correction: time each full inbound leg from wings level until reaching the fix. Note the difference between this time and one minute. Add or subtract this difference to the time of each outbound leg.
            - Note: do not time the first inbound leg as a part of the entry. This correction only applies to full inbound legs once established in the hold.
    - When holding, ensure the student is continuing with the radial scan.
    - Repeat procedures with different instructions until the student is proficient.
- Non-standard hold:
- <u>Student Tells, Student Does.</u>
    - Give the student instructions to join a Victor or Tango airway which leads to an intersection with another airway.
    - Give the student a non-standard holding instruction:
        - *Hold N of VALKA on T336, left turns, EFC 15 mins.*
    - Ensure that the student repeats and understands the instructions.
    - Turn on the bearing pointer and set its navigation source to the cross-airway. The fix occurs when the bearing pointer reaches the course of the cross-airway.
    - Ensure that the student can determine the correct entry procedure.

- Verify the completion of the 6 Ts (note that the order may change slightly depending on entry):
    - **T**urn - upon crossing fix, turn to the outbound heading.
    - **T**alk - advise ATC established in the hold.
    - **T**ime - start the timer at wings level - **there is no flag flip on a DME hold.**
    - **T**wist: twist the CDI to the inbound course **if necessary.**
        - Refer to intersection holding patterns in Lesson 314.
        - If **direct**, there will be no need to twist.
        - If **parallel**, twist to the reciprocal of the current CDI course.
    - **T**rack - after one minute or the holding distance, turn inbound and track the inbound course.
    - **T**hrottle - adjust throttle for 90 kts.
- Ensure the student is making proper wind and time corrections.
    - Wind correction: once established inbound for the first time, note the difference between the current heading and the ground track on the PFD. This difference x 3 should be applied into the direction of the wind to the outbound heading.
    - Time correction: time each full inbound leg from wings level until reaching the fix. Note the difference between this time and one minute. Add or subtract this difference to the time of each outbound leg.
        - Note: do not time the first inbound leg as a part of the entry. This correction only applies to full inbound legs once established in the hold.
- When holding, ensure the student is continuing with the radial scan.
- Repeat procedures with different instructions until the student is proficient.

<u>Post-Brief</u>

- Student Self Critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 317
- Assign Homework:
    - Assign holding practice assignment.
    - Study holding entries and procedures.

<u>Completion Standards</u>

The student should demonstrate the correct procedures in order to execute DME Fix and GPS holding patterns.